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(54) **PROCESSING METHODS AND PROCESSING APPARATUS**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

1,648,937 A 11/1927 Dickerson
1,989,406 A * 1/1935 Doolittle B01D 1/18
159/4.06

(Continued)

FOREIGN PATENT DOCUMENTS

CN 203065548 U 7/2013
JP 2011202193 B1 10/2011
WO 9306254 A1 4/1993

OTHER PUBLICATIONS

Daintith, John Martin, Elizabeth. Dictionary of Science, 2010, Oxford University Press, 6th Edition, p. 756 (Year: 2010).*

(Continued)

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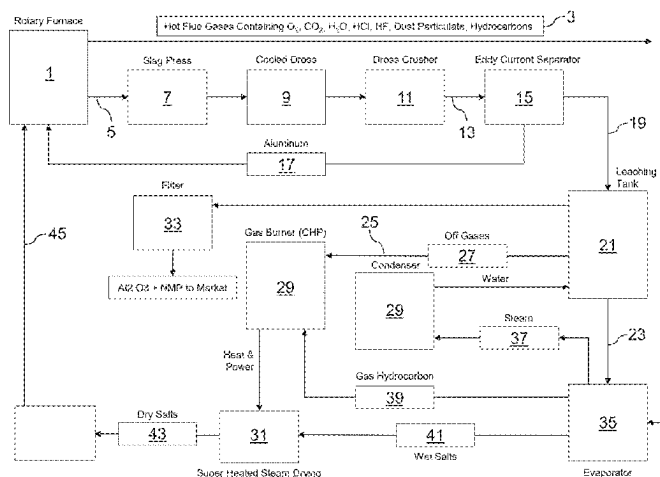
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(57) **ABSTRACT**

A method and apparatus for processing a material are provided, the material being the upper layer from a metal melting process, the material containing one or more salts, the material containing one or more metals, the salts and/or metals being recycled as a result of the method/apparatus. The method includes feeding the material to a leaching step; obtaining a leachate from the leaching step; feeding the leachate to a drying step or spray drying step; obtaining a solid from the drying step or spray drying step. Off gases from the leaching step are used to provide heat to the drying step. The drying step provides a product well suited to being turned into pellets for reuse.

7 Claims, 3 Drawing Sheets



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See application file for complete search history.
- (56) **References Cited**
U.S. PATENT DOCUMENTS
- | | | | | | |
|-----------|-----|---------|-------------------|-------------|--|
| 3,607,221 | A * | 9/1971 | Kibby | C22B 21/02 | |
| | | | | 75/10.21 | |
| 4,050,999 | A | 9/1977 | Johnson | | |
| 4,073,644 | A | 2/1978 | Papafingos et al. | | |
| 4,090,916 | A * | 5/1978 | Papafingos | C22B 21/062 | |
| | | | | 159/9.2 | |
| 4,386,956 | A | 6/1983 | Roth et al. | | |
| 4,477,373 | A * | 10/1984 | Grantham | B01J 10/005 | |
| | | | | 110/235 | |
- | | | | | | |
|--------------|------|---------|-----------------|--------------|--|
| 4,569,723 | A | 2/1986 | Lyon et al. | | |
| 5,102,453 | A | 4/1992 | Yerushalmi | | |
| 5,330,618 | A * | 7/1994 | Daniels | B01D 1/26 | |
| | | | | 423/205 | |
| 5,401,294 | A | 3/1995 | Kos et al. | | |
| 5,993,512 | A | 11/1999 | Pargeter et al. | | |
| 6,228,298 | B1 * | 5/2001 | Waatti | B01J 49/75 | |
| | | | | 264/112 | |
| 2002/0027304 | A1 * | 3/2002 | Robson | C22B 21/0069 | |
| | | | | 423/625 | |
| 2013/0075964 | A1 | 3/2013 | Peel et al. | | |
- OTHER PUBLICATIONS
- Advantages and Disadvantages of Spray Drying, Jul. 5, 2011 (Year: 2011).*
- Tsakiridis P.E. et al., "Aluminium salt slag characterization and utilization—A review," Journal of Hazardous Materials, Elsevier, Amsterdam, NL; vol. 217, Mar. 16, 2012 (Mar. 16, 2012), pp. 1-10, XP028412593.
- Zhang, Lifeng, "State of the Art in Aluminum Recycling from Aluminum Dross," Light Metals, TMS, Minerals, Metals and Materials Society / Aluminium Committee, US Mar. 12, 2006 (Mar. 12, 2006), pp. 931-936, XP009091955.
- ASM International Handbook Committee, "Powder Metal Technologies and Applications," Dec. 1, 1998 (Dec. 1, 1998), pp. 92-95, ASM International, USA, XP002733954.
- Delau, J. G. M., "Preparation of Ceramic Powders from Sulfate Solutions By Spray Drying and Roasting," Ceramic Bulletin, vol. 49, No. 6, Jan. 1, 1970 (Jan. 1, 1970), pp. 572-574, XP001268755.
- Schwartz, Harold. "Aluminum." ProKon Version 8.6 (Year: 1997). Examination Report from Indian Patent Office issued in corresponding PCT International Application No. GB2014053129 mailed on Dec. 26, 2019.
- * cited by examiner

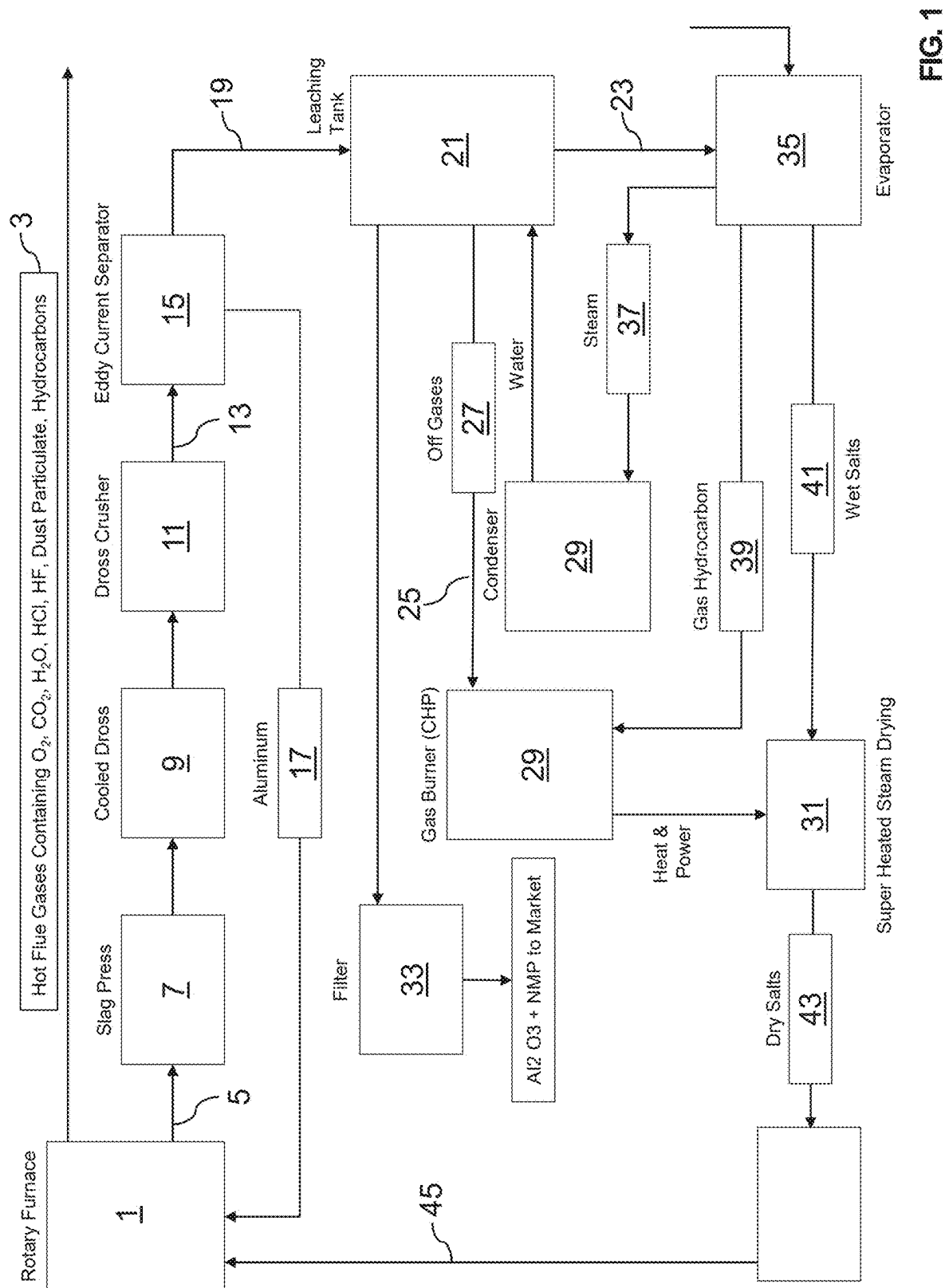


FIG. 1

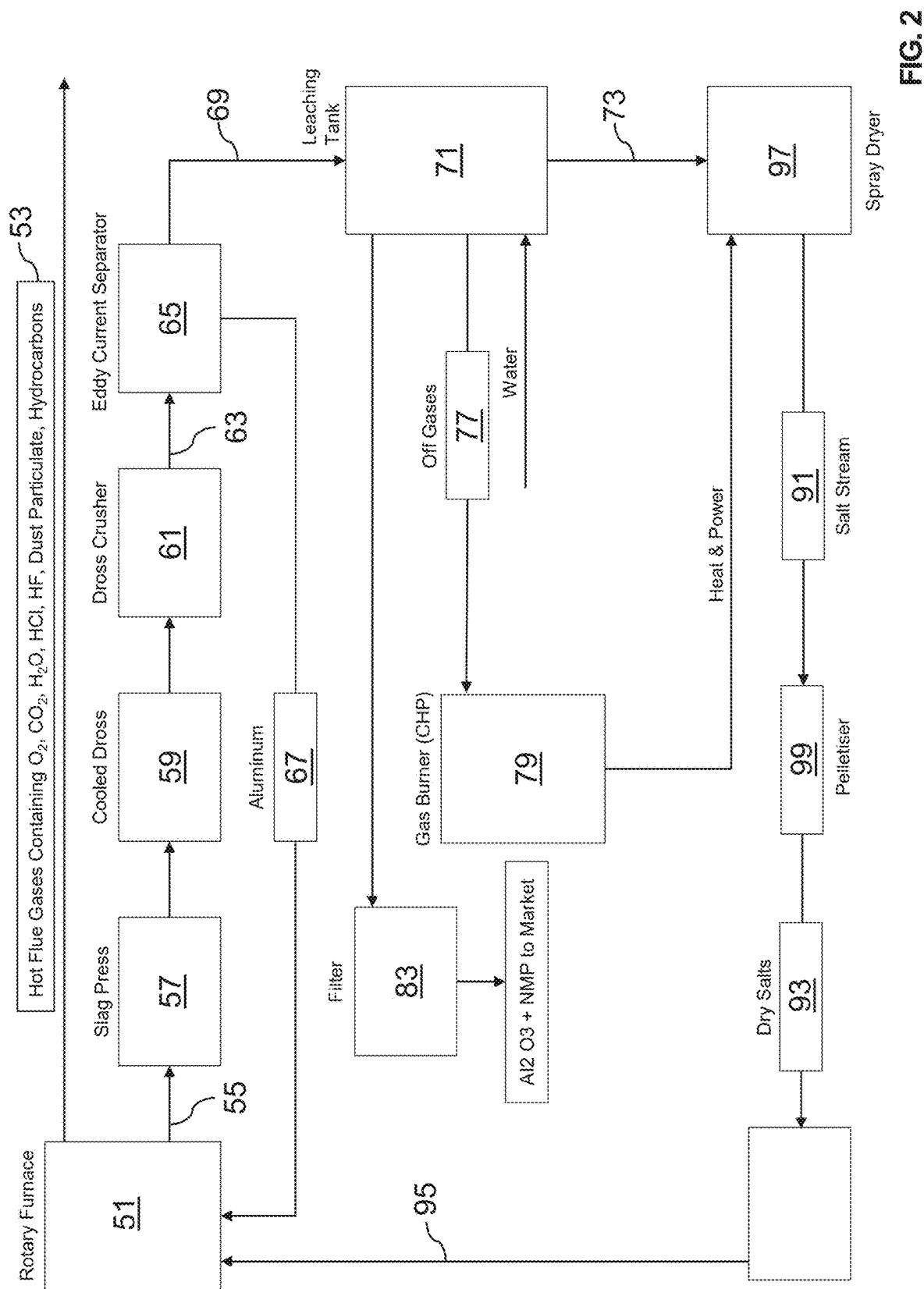


FIG. 2

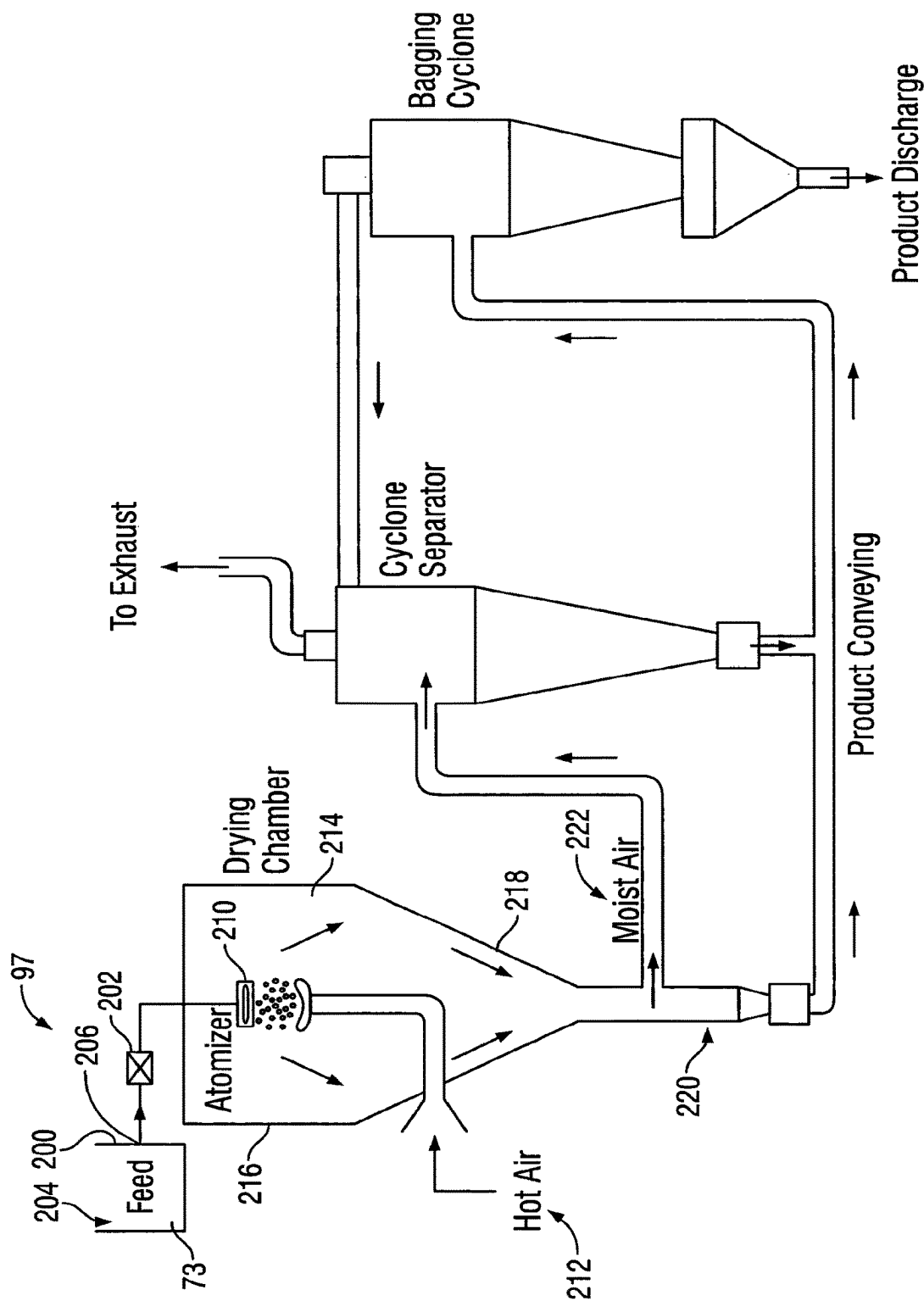


FIG. 3

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PROCESSING METHODS AND PROCESSING APPARATUS

This invention concerns improvements in and relating to processing methods and apparatus, particularly for processing slags to recover one or more useful components thereof.

Furnaces handling molten aluminium, for example, are provided with a salt barrier on top of the molten metal to reduce oxidation and/or promote the removal of contaminants from the molten metal. The salt barrier effectively becomes part of the slag as a result of the process, after which it is commonly referred to as a salt slag. There are advantages in processing this material once removed from the furnace. The advantages include the recovery and reuse of one or more of the components and/or a reduction in the hazards the slag presents.

Existing approaches tend to use centralised processing plants to take the cold slag and reprocess it. This causes transport issues in terms of cost and transport footprint and has a negative impact upon the subsequent processing and products.

The invention has amongst its possible aims to facilitate processing of the slags, economically at the same site it is produced at. The invention has amongst its possible aims to reduce costs involved in the processing. The invention has amongst its potential aims to give improved products, potentially in terms of their value and/or usefulness.

According to a first aspect there is provided a method of processing a material, the method including:

- a) feeding the material to a leaching step;
- b) obtaining a leachate from the leaching step;
- c) feeding the leachate to a drying step or spray drying step;
- d) obtaining a solid from the drying step or spray drying step.

The method may include the further step of compressing and/or moulding the solid into pellets.

The method may include 1) providing a material on the surface of a molten metal in a furnace, the furnace being at a first location; 2) removing the material from the furnace; 3) processing the material removed from the furnace at a second location, the second location being 5 km or less from the first location and the processing including one or more of steps a), b), c) or d). The method may include one or more of the processes at the second location being applied before the temperature of the material removed from the furnace has reduced below 100° C., preferably below 350° C., more preferably below 525° C. and yet more preferably below 760° C.

According to a second aspect there is provided apparatus for processing a material, the apparatus including:

- a) leaching apparatus;
- b) a feeder providing material to the leaching apparatus;
- c) an outlet for leachate from the leaching apparatus;
- d) a dryer or spray dryer;
- e) the outlet being fluidly connected to the spray dryer;
- f) an outlet from the dryer or spray dryer for solids.

The apparatus may include the compression and/or moulding apparatus for producing pellets and/or a feeder for providing a solid obtained from the leachate to the compression and/or moulding apparatus.

The apparatus may include 1) a furnace, the furnace being at a first location; 2) a feeder for introducing a material to the surface of a molten metal in the furnace; 3) one or more processors for the material removed from the furnace, the one or more processors being provided at a second location, the second location being 5 km or less from the first location

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and the one or more processors including one or more of the processors in one or more of features a), b), c), d), e) or f).

According to a third aspect there is provided a method of processing a material, the method including

- a) feeding the material to a leaching step;
- b) obtaining a leachate from the leaching step;
- c) obtaining a solid from the leachate;
- d) compressing and/or moulding the solid into pellets.

The method may include feeding the leachate to a drying step or spray drying step, preferably to obtain the solid for the compressing and/or moulding step.

The method may include 1) providing a material on the surface of a molten metal in a furnace, the furnace being at a first location; 2) removing the material from the furnace; 3) processing the material removed from the furnace at a second location, the second location being 5 km or less from the first location and the processing including one or more of steps a), b), c) or d). The method may include one or more of the processes at the second location being applied before the temperature of the material removed from the furnace has reduced below 100° C., preferably below 350° C., more preferably below 525° C. and yet more preferably below 760° C.

According to a fourth aspect there is provided apparatus for processing a material, the apparatus including:

- a) leaching apparatus;
- b) a feeder providing material to the leaching apparatus;
- c) an outlet for leachate from the leaching apparatus;
- d) compression and/or moulding apparatus for producing pellets;
- e) a feeder for providing a solid obtained from the leachate to the compression and/or moulding apparatus.

The apparatus may include a dryer or spray dryer and/or the outlet being fluidly connected to the dryer or spray dryer and/or an outlet from the dryer or spray dryer for solids.

The apparatus may include 1) a furnace, the furnace being at a first location; 2) a feeder for introducing a material to the surface of a molten metal in the furnace; 3) one or more processors for the material removed from the furnace, the one or more processors being provided at a second location, the second location being 5 km or less from the first location and the one or more processors including one or more of the processors in one or more of features a), b), c), d) or e).

According to a fifth aspect there is provided a method of processing a material, the method including

- a) providing a material on the surface of a molten metal in a furnace, the furnace being at a first location;
- b) removing the material from the furnace;
- c) processing the material removed from the furnace at a second location, the second location being 5 km or less from the first location and the processing including one or more steps selected from:
 - 1) reduction of the size of a mass of the material;
 - 2) leaching the material to provide a leachate;
 - 3) drying or spray drying one or more components, such as a leachate, to provide a solid;
 - 4) compressing or moulding a solid, for instance the solid obtained from a leachate, to provide pellets.

The method may include one or more of the processes at the second location being applied before the temperature of the material removed from the furnace has reduced below 100° C., preferably below 350° C., more preferably below 525° C. and yet more preferably below 760° C.

The method may include feeding the leachate to a drying step or spray drying step, preferably to obtain the solid for the compressing and/or moulding step.

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The method may include the further step of compressing and/or moulding the solid into pellets.

According to a sixth aspect there is provided apparatus for processing a material, the apparatus including

- a) a furnace, the furnace being at a first location;
- b) a feeder for introducing a material to the surface of a molten metal in the furnace;
- c) one or more processors for the material removed from the furnace, the one or more processors being provided at a second location, the second location being 5 km or less from the first location and the one or more processors including one or more processors selected from:
 - 1) a breaker for reducing of the size of a mass of the material;
 - 2) leaching apparatus;
 - 3) drying or spray drying step apparatus;
 - 4) compression or moulding apparatus for a solid, for instance the solid obtained from a leachate from the leaching apparatus, to provide pellets.

The first and/or second and/or third and/or fourth and/or fifth and/or sixth aspects of the invention may include any of the features, options or possibilities set out elsewhere within this document, including from amongst the following.

The method of processing may be processing to separate one or more salts from one or more metals. For instance sodium chloride and/or sodium chlorate and/or potassium chloride and/or potassium chlorate may be separated, particularly from aluminium and/or ferrous metals.

The method of processing may be processing to separate one or more salts from one or more non-metals. For instance sodium chloride and/or sodium chlorate and/or potassium chloride and/or potassium chlorate may be separated, particularly from aluminium oxides and/or other insoluble non-metallic compounds.

The method may provide for the recycling of one or more metals, for instance aluminium. The method may provide for the recycling of one or more salts, for instance sodium chloride and/or potassium chloride.

The material may be a slag. The material may be the upper layer from a metal melting process. The material may be a layer involved in recycling aluminium. The material may contain one or more salts, for instance sodium chloride and/or sodium chlorate and/or potassium chloride and/or potassium chlorate. The material may contain one or more metals, for instance aluminium, and/or metal oxides, for instance aluminium oxide. The material may contain 10-35% aluminium oxide, 25-60% sodium chloride, 10-35% potassium chloride, 2-10% metallic aluminium and impurities, with the combination totalling 100%, for instance 15-30% aluminium oxide, 30-55% sodium chloride, 15-30% potassium chloride, 5-7% metallic aluminium and impurities, with the combination totalling 100%. The impurities may include carbides, nitrides, sulphides and/or phosphides.

The method of providing a material on the surface of a molten metal in a furnace may include feeding the material onto the surface of the molten metal, preferably from above the molten metal. The material may be fed in batches. The material may be fed continuously.

The molten metal is preferably aluminium.

The furnace may be a rotary furnace or other furnace type.

The first location may be the location at which the furnace is provided and/or at which the material is stored after removal from the furnace and/or after being compressed.

The material may be a slag. The material may be a compressed slag produced by compressing a slag. The slag may be compressed in a slag press. The slag may be

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compressed to remove molten aluminium from the slag. The material may contain one or more salts, for instance sodium chloride and/or sodium chlorate and/or potassium chloride and/or potassium chlorate. The material may contain one or more metals, for instance aluminium, and/or metal oxides, for instance aluminium oxide, ideally at a lower level than in the slag.

The compressed slag may be of a lower temperature than the slag. The compressed slag may have a temperature of less than 500° C. after compression.

The compressed slag may be allowed to cool. Preferably the compressed slag is not allowed to cool below 100° C., preferably below 350° C., more preferably below 525° C. and yet more preferably below 760° C.

The compressed slag may be allowed to cool at the first location.

The second location may process less than 20,000 tonnes of material per year, potentially less than 15,000 tonnes and even less than 10,000 tonnes of material per year.

The processing of the material removed from the furnace which is provided at a second location may include one or more of the steps subsequent to the removal of the material from the furnace and more preferably subsequent to the compression of the material, for instance to form the compressed slag.

The second location is 5 km or less from the first location, and preferably is less than 2 km from the first location, more preferably is less than 1 km from the first location and ideally is less than 500 m from the first location. The first location and the second location are preferably owned and/or operated by the same legal entity. The furnace is preferably fed with recycled aluminium.

The compressed slag preferably is not allowed to cool below 100° C., preferably below 350° C., more preferably below 525° C. and yet more preferably below 760° C. before one or more process step are conducted at the second location.

The method may include a size reduction step. The size reduction step may reduce one or more dimensions of the material and/or increase the number of discrete parts in which the material is present. The size reduction step may include a plurality of size reduction steps. The size reduction step may include a crushing step. The size reduction step may include a grinding step. The size reduction step may include one or more sizing steps, for instance to separate the material into one or more fractions according to size.

The method may include a ferrous and non-ferrous metal separation step. The separation may be made by an eddy current separator. The method may include a metal and non-metal separation step. The separation may be made by an eddy current separator and/or vibrating table and/or compressed air separator. Aluminium extracted by one or both of these separations may be returned to the furnace.

The method may include feeding the material to a leaching step by introducing the material to a leaching tank. The leaching step may be provided with one or more other feeds, for instance water.

The method may include obtaining a leachate from the leaching step or steps.

The leaching step may be a multiple pass leaching process where multiple passes through one leaching tank are provided and/or a multiple stage leaching process where passage through a number of leaching tanks is provided. The method may include obtaining the leachate from the leaching step or steps by extracting the leachate from a leaching tank, preferably from the bottom thereof. The leaching step

may be provided with one or more other outlet streams, for instance an off gas stream and/or non-dissolved solids stream.

The off gas stream may be further processed, for instance by being combusted, ideally with heat and/or power generated by the combustion being used in one or more of the method steps and ideally the drying step or spray drying step.

The non-dissolved solids stream may be further processed, for instance in a solid liquid separator such as a filter. The separated solids may form a by-product, ideally which has a market value. The separated liquids may be returned to the leaching step and/or may be fed to the spray drying step.

The method may include feeding the leachate to a drying step or spray drying step, for instance via one or more buffering or storage containers. The leachate may be pumped to the drying step or spray drying step, and particularly one or multiple nozzles therein.

Particularly in relation to spray drying, one or more gases, for instance air, may be pumped to the drying step, particularly the nozzle/nozzles therein. The nozzle may co-inject the gas and leachate. The nozzle may act as an atomiser of the fluid leachate

The drying step may use and/or the dryer may be a falling film evaporator. The leachate may flow down over one or more surfaces provided in the dryer, for instance the surfaces of one or more tubes. The leachate may flow as a thin film on the surface. One or more or all of the surfaces may be heated. The whole or a part of one or more or all of the surfaces may be heated. Gases, for instance air, may pass over the leachate, preferably moving in a direction different to, for instance opposing, the direction of movement of the leachate.

The method may include feeding hot gas, preferably hot air, to the drying step or spray drying step. The hot gas may be provided in a parallel or counter flow direction to the introduction of the leachate.

The gases may be the hot gases from the off gas stream processing and/or may possess energy generated by the off gas stream processing, particularly in terms of their temperature.

One or more additional components, for instance water or steam, may be added to the drying step, for instance co-injected with the gases. One or more additional components may be added to control the pressure and/or temperature and/or chemical composition of the feed to the drying step. The hot gas may be generated by a heating step, for instance by electrical heating and/or heating exchanger system.

The atomised leachate is preferably dried by contact with the hot gas. The atomised leachate is preferably solidified by the hot gas.

The method may include obtaining a solid from the drying step or spray drying step, for instance by separating the solid and the gas in the drying step or spray drying step. The solid may be collected at the bottom of the drying step or spray drying step. The solid may be removed in batches or continuously. One or more subsequent separations of solids from gas may be provided for the gas outlet stream, for instance using bag filters and/or cyclone separators. The gas outlet stream, preferably after solids separation, may be used to agitate and/or heat in one or more of the method steps, particularly in the drying step or spray drying step.

The method may include obtaining the solid from the leachate by using the drying step or spray drying step.

The method may include compressing and/or moulding the solid into pellets in one or more further steps. The one or more further steps may include a pelletising step. The

compressing and/or moulding may be provided without adding any further components to the solid. In particular no binders or moisture may be added.

The compressed and/or moulded solids, for instance pellets, may be collected and/or stored, for instance for subsequent use. The compressed and/or moulded solids, for instance pellets, may be fed to a furnace, ideally the furnace at the first location. The compressed and/or moulded solids, for instance pellets, may be fed to a furnace, ideally the furnace at the first location, to provide a material on the surface of a molten metal in a furnace.

The compressed and/or moulded solids, for instance pellets, may be produced to a controlled size.

The compressed and/or moulded solids, for instance pellets, may be produced to a controlled size distribution, for instance such that less than 5%, more preferably less than 2%, of the pelletised material by weight is less than 149 microns in diameter, for instance considered as the largest linear dimension.

The compressed and/or moulded solids, for instance pellets, may be produced to a controlled moisture level, for instance less than 0.5 wt % or preferably less than 0.2 wt %.

The compressed and/or moulded solids, for instance pellets, may be produced with a controlled ratio of sodium chloride to potassium chloride, for instance a predetermined ratio $\pm 5\%$. The ratio may be between 4:1 and 1:1 sodium chloride to potassium chloride.

One or more batches of compressed and/or moulded solids, for instance pellets, having a first set of characteristics may be blended with one or more other batches of compressed and/or moulded solids, for instance pellets, having one or more different characteristics.

Various embodiments of the invention will now be described, by way of example only, and with reference to the accompanying drawings in which:

FIG. 1 is a flow diagram for a process for the recovery of salt from an aluminium recycling process, including improvements according to the invention;

FIG. 2 is a flow diagram for a process according to an embodiment of the invention;

FIG. 3 is a schematic illustration of a spray dryer for use in the invention.

There are various problems with the large scale centralised process. These include:

The transport costs in taking the compressed slag 9 to a suitable centralised plant for processing;

The significant regulatory issues (particularly if country borders are crossed) with such transportation;

The safety risk from accidents and the like involved in such transportation;

Having to use a centralised process plant, outside of the salt producer's control;

The increased energy requirements and difficulty in achieving size reduction of the cold compressed slag;

The variable nature of the feed slag, due to its containing different salt levels, different slag proportions, different contaminants and the like as it arrives from various different aluminium recyclers, giving process control issues;

The variable nature of feed slag impacting upon the dry salt produced (in terms of its chemical make up, moisture content and size distribution) meaning the product is not well controlled or defined and so impairs it as a good salt feed for the furnace.

In FIG. 1, a rotary furnace 1 is used to recycle aluminium, such as scrap which is contaminated with paint, lacquers and

the like. The furnace **1** generates an aluminium product stream (not shown) together with an off gas stream **3** and a slag stream **5**.

The off gas stream **3** is subjected to suitable processing to recover/use or render safe the gases it contains.

To prevent oxidation of the aluminium and to provide other benefits, a salt is added to the top of the melt in the furnace **1** to form a barrier. The salt, together with some aluminium, forms the slag and hence the slag stream **5**. The salt is mainly sodium chloride, but with potassium chloride too and other materials. A typical salt slag (also known as aluminium salt cake) produced by the secondary aluminium industry may contain 15-30% aluminium oxide, 30-55% sodium chloride, 15-30% potassium chloride, 5-7% other materials (such as aluminium and impurities including carbides, nitrides, sulphides and phosphides etc).

After processing, the aluminium is drained from the furnace **1** and the slag is the main material remaining. The slag is formed mainly of the salt, together with the removed contaminants and other components picked up in the melting process.

When the slag stream **5** is removed from the furnace **1** it is generally emptied into bins (not shown) and then fed to a slag press **7** to allow for mechanically assisted agglomeration of the aluminium from the slag. The aluminium is generally returned to the furnace **1** after separation. The compressed material from the slag press **7** is classified as compressed slag **9**.

The compressed slag **9** is allowed to cool. The compressed slag **9** still contains a material amount of aluminium and a large amount of salt. The compressed slag is classified as a hazardous material, with respect to environmental and other laws, in many countries and so requires further consideration.

One option is to transport the compressed slag **9** to a suitable plant for the recovery of the aluminium and treatment to recover the salt. This involves road transportation of the compressed slag **9** from the site of the furnace **1** where the compressed slag **9** is generated, to the distant processing plant. This involves significant transport costs, significant regulatory issues (particularly if country borders are crossed) and a safety risk from accidents and the like.

The typical process plants are large installations which provide centralised treatment of the compressed slag **9** arising from various aluminium melting plants.

The compressed slag **9** is cold by the time processing starts and is fed to a salt crusher **11** to reduce the compressed slag **9** in size. The cold nature of the compressed slag **9** presents difficulties and increased energy requirements for the size reduction steps. For instance, pneumatic breakers may be needed to initially break up the compressed slag **9** into smaller lumps. The size reduction may include the illustrated salt crusher **11**, together with size based separation steps, such as sieves.

The size reduced salt stream **13** is then fed to an eddy current separator **15**. This generates a ferrous stream (not shown) which is recycled, an aluminium stream **17** which may be collected and sold and/or may be returned to the furnace **1** and a cleaned salt stream **19**.

The cleaned salt stream **19** is fed to a leaching tank **21**, together with a water stream. The leaching process dissolves the salt content of the cleaned salt stream **19**, ionises the sodium chlorate and so provides a dissolved salt stream **23**. The remaining solids, generally aluminium oxides and non-metallic particles form suspended solid stream **25**. The leaching off gas stream **27** is fed to a combined heat and power unit **29** where it is burnt to provide power and heat to

the overall process, in particular to the steam dryer **31**. Typical gases include hydrogen, methane, and ammonia. Phosphine and hydrogen sulphide may also be present.

The suspended solid stream **25** is fed to a filter **33** to separate the solids from the liquid. The solids are a useful by product of the process which can be sold. The liquids are returned (not shown) to the leaching tank **21**.

The dissolved salt stream **23** is passed to an evaporator **35** to give rise to a steam stream **37**, a gas hydrocarbon stream **39** (which also goes to the combined heat and power unit **29**) and a wet salt stream **41**.

The steam stream **37** is fed to a condenser **43** and the resulting condensed water stream is fed back to the leaching tank **21**.

The wet salt stream **41** is taken to a super heated steam dryer **31**. This produces a steam/water vapour stream (not shown) which is returned to the leaching tank **21** and the products stream, the dry salt stream **43**. The dry salt stream **43** can provide salt feed **45** for the furnace **1**.

The present invention attempts to address these issues and/or provide improvements in them as a result of the process approach and changes made.

Referring to FIG. 2, the overall process of an embodiment of the invention is provided. Once again, the rotary furnace **51** is used to recycle aluminium and the furnace **51** generates an aluminium product stream (not shown) together with an off gas stream **53** and a slag stream **55**.

Features in common with the approach of FIG. 1 are not repeated, but are still applicable to the FIG. 2 embodiment. For instance, the off gas stream **53** is subjected to suitable processing to recover/use or render safe the gases it contains.

The slag stream **55** is removed from the furnace **51** it is then fed to a slag press **57**. The compressed material from the slag press **57** is again the compressed slag **59**.

In this embodiment of the invention, the compressed slag **59** is processed at the same site as it is generated at. This is possible because of the lower capital cost and lower unit sizes viable for the overall process selected in this embodiment when compared with the prior approach. A typical localised plant might be configured to handle 1 tonne or less per hour of salt, whereas centralised plants may handle 5 or more tonnes per hour.

A first benefit arising from this is that the compressed slag **59** is only allowed to cool to the desired extent. That cooling still aids handling and reduces the risk of the hot aluminium reacting further, but the compressed slag **59** is not allowed to cool to an extent that would cause negative impacts in the subsequent size reduction stage.

The compressed slag **59** is fed to a slag crusher **61** to reduce the compressed slag **59** in size. However, as the compressed slag **59** is not cold and hard, the size reduction is easier to achieve and the size distribution resulting is more readily controlled. This gives a better size distribution and/or reduced amount of fine material, for instance sub 1 mm material. This is significant in improving the saleability of the solids recovered from the suspended solids stream **75**.

After size reduction, the size reduced slag stream **63** is then fed to an eddy current separator **65**. This generates a ferrous stream (not shown) which is recycled, an aluminium stream **67** which is returned to the furnace **1** and a cleaned salt stream **69**.

The cleaned salt stream **69** is fed to a leaching tank **71**, together with a water stream **73**. The same general leaching process is provided.

The known nature of the slag **59**, because the aluminium recycling plant operator and the slag processing plant opera-

tor is the same, mean that the feed material to the leaching tank **71** is better known and more consistent.

The dissolved salt stream **73**, the suspended solid stream **75** and the leaching off gas stream **77** arise as before. The leaching off gas stream **77**, however, is now made use of to increase the energy efficiency of the overall system. The leaching off gas stream **77** is fed to a combined heat and power unit **79** where it is burnt to provide power and heat to the overall process. In particular, the power can be made of throughout the process to drive one or more processes and the heat can be made use of in the spray dryer **97**. Typical gases present include hydrogen, methane, and ammonia.

The suspended solid stream **75** is fed to a filter **83** to separate the solids from the liquid. The solids are a useful by product of the process which can be sold and with improved properties and hence value, as mentioned above.

The dissolved salt stream **73** is then subjected to different processing steps than before.

The dissolved salt stream **73** passed to the spray dryer **97**, described in more detail below. The spray dryer benefits from heat recycled from the furnace **51** and/or from the combined heat and power unit **59**. The salt stream **91** arising is much drier than from the evaporator in the FIG. 1 form.

In an embodiment not shown, the dissolved salt stream **73** may be passed to an evaporator to give rise to a steam stream, a gas hydrocarbon stream (which also goes to the combined heat and power unit **79**) and a wet salt stream **73** which then passes to the spray dryer **97** described above. An issue with the salt stream **91** is that the particle size is relatively small. However, the applicant has identified that this issue can be address by feeding the salt stream **91** to a pelletiser **99**, described in more detail below. Furthermore, the applicant has identified that the salt stream properties and size are ideally suited to pelletisation without the need for further treatment or additives, such as binders.

As the product stream, the dry salt stream **93** is now pelletized, this allows for the optimisation of the physical properties, such as the size of the pellets, for the subsequent reuse of the dry salt stream **93** as a salt feed **95** for the furnace **51**.

In addition, the pellets are known to be of a standard content and so offer more controlled feeds to the furnace **51**, for instance in terms of the ratio of sodium chloride to potassium chloride present (which is used to control the sodium chloride melting temperature). The blending of different pellets is also possible to provide other variation in the feed to the furnace **51**.

Finally, the pellets are also more suited to storage, without moisture take up, for instance, than the dry salt product of the FIG. 1 form.

As illustrated in FIG. 3, the spray dryer **97** receives the dissolved salt stream **73** via a balance tank **200**, with the flow into the tank controlled by a valve **202** to maintain the solution level. Low to open and high to close level probes are used to operate the valve **202**. A water inlet **204** is also provided to the balance tank **200**. The outlet **206** supplies the dissolved salt stream **73** to a peristaltic pump which pumps the dissolved salt stream **73** to the atomiser **210**.

The atomiser **210** also receives filtered air.

The atomiser **210** makes use of a two fluid nozzle atomiser offering co-current mode, but rotary atomisers and two fluid atomisers in fountain mode are possible too. The high velocity air encounters the low velocity fluid to disintegrate that into droplets.

The atomiser **210** is positioned in the upper part of the drying chamber **214**, with the drying chamber **214** being formed of a cylindrical upper part **216** and conical lower part

218. The bottom of the conical part **218** provides the powder outlet **220** and moist air side outlet **222**.

The atomised fluid directly encounters the hot air which has passed through the heating stage **212**. The heating stage **212** includes the electrical heat source and baffled air distributors to control the flow pattern and heat transfer in the air.

Upon entering the drying chamber **214** through the atomiser **210**, the dissolved salt stream **73** comes into intimate contact with the hot air. This provides for the drying in the hot air stream. Away from the atomiser **210** the drying chamber **214** allows the separation of the dry powder from the hot air, with both passing to their respective outlets.

The dry powder is conveyed away by a fluidised ejector. Cooling and packing of the powder can then be provided for.

The separated hot air enters an air stream going to atmosphere which is provided with bag filters to remove any remaining powder, with cyclones offering an alternative form of filtering.

The following table defines some typical operating characteristics of a suitable spray dryer.

TABLE 1

Characteristic	Value	Units
Processed material	Chloride based salts	
Average residence time in dryer	11	s
Material feed rate	100	Kg/hr
Feed solids content	30	%
Rate of powder production	30	Kg/hr
Evaporation rate of dryer	70	Kg/hr
Residual moisture in product	1	%
Drying air inlet temperature	190	° C.
Drying air outlet temperature	97	° C.
Powder temperature	85	° C.
Mass flow rate of drying air	2454	Kg/hr
Heat input to dryer	103905	Kcal/hr
Heat input to dryer	120	KW
Operational running load	20	HP
Electrical supply	440-3-50	VV-ph-Hz
Instrumentation	24	V

The spray drying product has closely controlled particle size distribution, residual moisture content, bulk density and morphology

In terms of the results achieved, the variation in moisture content achieved with varying conditions is detailed in Table 2.

TABLE 2

Trial	Inlet Temp ° C.	Outlet Temp ° C.	Feed Rate L/hr	Salt Conc wt %	NaCl:KCl	Moisture wt %
1	150	90	1.2	24.2	66:34	0.27
2	195	102	2	24.2	66:34	1.05
3	150	90	1.2	24.6	50:50	0.31
4	195	100	2	24.6	50:50	1.64
5	150	90	1.2	16.7	66:34	0.32
6	196	102	2	16.7	66:34	
7	150	90	1.2	17.0	49:51	1.18
8	197	99	2	17.0	49:51	1.40

The particles generated, when measured using SEM imaging, are generally around 5 to 30 microns in diameter and are blocky crystals in appearance. Optical microscope imaging also confirmed that the particles tend to form agglomerates of 120 to 2500 microns in diameter.

The pelletiser **99** is important in increasing the size of the product to a size at which it is not prone to airborne transportation out of the furnace.

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Roll based compactors were used to examine the suitability of the powder to pelletizing. The powder was fed using gravity through two counter rotating rollers. Friction between the material and the roller surfaces brought the powder into the narrow space between the rolls and the high loads present gave good compaction of the powder. The powder held together in the compacted form without the need for any binders, water or other additives to be present.

Roll based pelletisers, where the rolls bear opposing halves of the recessed spaces where the pellets form where then used to form pellets successfully.

The technique readily achieves moisture contents below 0.2 wt % and allows less than 2% of the material to be below 149 microns in diameter (so minimising the amount of fines present).

The invention claimed is:

1. A system for processing a material containing one or more salts and one or more metals, the system including:

a leaching tank having:

an inlet configured to provide material to the leaching tank; and

a first outlet configured to transfer leachate from the leaching tank;

a dryer having:

an inlet directly coupled to the outlet of the leaching tank;

an outlet configured to transfer solids from the dryer;

a first feeder configured to provide off gas from the leaching tank to an off gas stream processor; and

a second feeder configured to provide hot gas to the dryer, wherein the hot gas is provided to the second feeder by

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the off gas stream processor and/or possesses energy generated by the off gas stream processor.

2. The system of claim 1, wherein the leaching tank includes a second outlet configured to transfer a suspended solid stream, that is separate from the leachate, from the leaching tank.

3. The system of claim 1, wherein the dryer is a spray dryer.

4. The system of claim 2, further comprising:

A leachate flow path configured to provide the leachate from the first outlet of the leaching tank to the dryer; and

a suspended solid flow path configured to provide the suspended solid stream from the second outlet of the leaching tank to a filter; and

wherein the leachate is not sent through the suspended solid flow path.

5. The system of claim 1, wherein the outlet of the dryer includes a first discharge configured to transfer a salt stream from the dryer and a second discharge configured to transfer hot air from the dryer, where the salt stream is in the form of a dry powder.

6. The system of claim 5, further comprising a pelletiser configured to receive the salt stream from the dryer and apply a force to the salt stream to form pellets.

7. The system of claim 1, wherein the off gas stream processor is configured to generate power and provide at least a portion of the generated power to the dryer in addition to the hot gas.

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